In-Class Assignment: Naive-Bayes

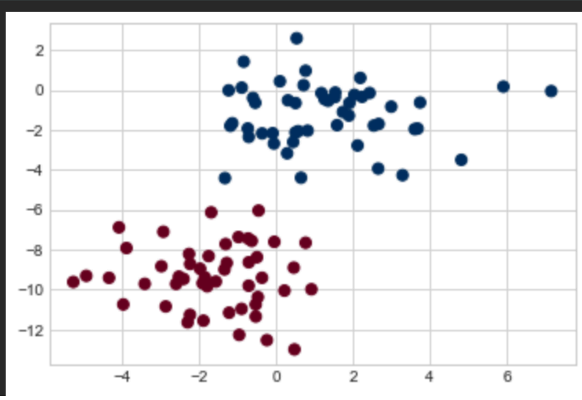
Please answer the following questions.
please refer to the [Naive-Bayes.ipynb - Colab]

Question 1:
What is the primary characteristic of Naive Bayes classifiers that makes them suitable for very high-dimensional datasets and useful as a quick-and-dirty baseline?

Question 2:
In the context of Gaussian Naive Bayes, what is the simplifying assumption made about the data from each label?

Question 3:
In the make_blobs function, what visual change would you expect if you modify the centers parameter from 2 to 5 and run the cell?

```python
from sklearn.datasets import make_blobs
X, y = make_blobs(100, 2, centers=2, random_state=2, cluster_std=1.5)
plt.scatter(X[:, 0], X[:, 1], c=y, s=50, cmap='RdBu');
```



Question 4:
In the scatter plot generation cell, if you change s=20 to s=200, and alpha=0.1 to alpha=0.5 for Xnew points, what visual difference would you observe?

```python
plt.scatter(X[:, 0], X[:, 1], c=y, s=50, cmap='RdBu')
lim = plt.axis()
plt.scatter(Xnew[:, 0], Xnew[:, 1], c=ynew, s=20, cmap='RdBu', alpha=0.1)
plt.axis(lim);
```